Promoting Inclusion and Peer Participation Through Assessment-Based Intervention

Kwang-Sun Cho Blair
University of South Florida
John Umbreit
University of Arizona
Glen Dunlap
University of South Florida
Gilsoon Jung
Moksang Elementary School

In the current investigation, the processes of functional assessment and function-based intervention were used to resolve the severe challenging behaviors of a boy with autism and mental retardation in an inclusive kindergarten in South Korea. A multicomponent intervention was developed in collaboration with classroom personnel and was implemented entirely by the teacher and an aide in the context of a multiple-baseline-across-activities experimental design. Results were empirical validation of hypotheses derived from the functional assessment, as well as lower levels of challenging behaviors and increased rates of appropriate behaviors associated with the intervention. Positive interactions with a designated classroom peer and with the teacher also increased. The findings are discussed as contributions to the growing literature on functional assessment and function-based supports and the importance of promoting successful inclusive experiences for young children with disabilities.

Challenging behaviors present the single biggest obstacle to including young children with disabilities in typical early education programs (Conroy, Dunlap, Clarke, & Alter, 2005). If challenging behaviors are not reduced and replaced with more appropriate behavior, students who display them are likely to be removed from an inclusive class and placed in a more restrictive environment (Carta et al., 1994; Palfrey, Singer, Walker, & Butler, 1987).

The most effective way to reduce challenging behaviors and promote prosocial behaviors is through a function-based approach (J. J. Fox & Gable, 2004; Heckaman, Conroy, Fox, & Chait, 2000; Sasso, Conroy, Stichter, & Fox, 2001). This process begins with a functional behavioral assessment (FBA) that identifies the consequences that reinforce a problem behavior and the events that predict when it will occur (Sugai et al., 2000). Experts then design interventions to address these variables.

Researchers have reported some success in using function-based intervention as a tool to maintain the inclusive placements of young children with disabilities. For example, authors have reported successful interventions with young children who were at risk for behavior disorders (Umbreit & Blair, 1997), as well as with those diagnosed with emotional and behavioral disorders (Blair, Umbreit, & Bos, 1999), mild retardation (Umbreit, 1996), and severe retardation (Blair, Liaupsin, Umbreit, & Kweon, 2006). Although studies have reported positive results, including improvements in student behavior and positive teacher interaction, only one (Duda, Dunlap, Fox, Lentini, & Clarke, 2004) has examined more than one activity during the school day. In addition, many of the studies on function-based intervention with young children with disabilities (e.g., Cunningham & O'Neill, 2000; Ellis & Magee, 1999; Hagopian, Wilson, & Wilder, 2001; Volkert, Lerman, & Vorndran, 2005) were conducted in therapy rooms or segregated classrooms based on assessments that employed analog rather than naturalistic methods. Overall, there are few successful demonstrations of the approach with young children in natural inclusive settings (Duda et al., 2004).

Many young children with disabilities have difficulties relating to peers, often displaying problem behaviors that elicit maladaptive social responses from their peers (Whalen & Henker, 1985). Therefore, researchers have paid attention to social skills interventions to prevent future problems (L. Fox, Dunlap, & Cushing, 2002; Strain & Cordisco, 1994). At the same time, researchers interested in function-based intervention for young children with disabilities and challenging behavior have paid little attention to the issue of peer interactions when designing interventions (Blair, Umbreit, & Eck, 2000). Developing intervention strategies that facilitate positive peer interaction and investigating the collateral effects of the intervention on peers should be a major target area in supporting young children with challenging behavior in inclusive classrooms.

The cultural appropriateness of interventions and practices is also a concern in considerations of approaches to challenging behavior. The current study was

Address: Kwang-Sun Cho Blair, University of South Florida, Child and Family Studies, FMHI/MHC 2113A, 13301 Bruce B. Downs Blvd., Tampa, FL 33612-3807; e-mail: kblair@fmhi.usf.edu

conducted in South Korea, where corporal punishment in schools continues to be lawful. According to an official government report, 7,536 out of 10,381 schools allow teachers to administer corporal punishment ("7 in 10 Schools Allow Corporal Punishment," 2003). Thus, educators still widely use aversive treatments for students with disabilities and challenging behavior who receive special education services, even in inclusive settings (S. Park, Park, & Oh, 2001). Researchers have found that problem behaviors of students with disabilities, in particular students with severe disabilities, are one of the most significant challenges in educational settings, and methods of managing problem behavior are the first priority for inservice training by special educators (E. Park & Park, 2001).

A growing number of researchers in South Korea are now interested in FBA and positive behavior support in addressing the challenging behaviors of students with disabilities. Nevertheless, only a few studies conducted in Korea have reported the results of interventions using this approach (Blair et al., 2006; S. Park et al., 2001). Studies are needed to evaluate the efficacy of FBA and function-based intervention that facilitates collaboration between general and special education staff and their active involvement in the entire process in supporting students with disabilities in South Korea.

A critical aspect of function-based intervention in natural classroom settings is the teacher's skill in conducting FBAs and developing and implementing intervention strategies (Reichle et al., 1996; Scott & Nelson, 1999). Therefore, providing teacher training and technical assistance are important elements in helping teachers become active natural agents in developing and implementing function-based interventions (Crone, Hawken, & Bergstrom, 2007; Dunlap et al., 2000). Studies that focus on building the capacity of teachers and demonstrating their proficiency in implementing interventions with high fidelity would provide valuable information about the applicability of the function-based intervention in inclusive settings in South Korea.

The primary purpose of this study was to examine further the usefulness of function-based intervention as a tool in supporting inclusive placements. The study was conducted in two phases. In Phase 1, data from an FBA were used to identify components of a function-based intervention that were tested directly within ongoing classroom activities. In Phase 2, a multicomponent intervention was derived from the FBA and implemented over time in multiple classroom activities. If an intervention improves the behavior of teachers and peers, in addition to the student in question, the prospects for successful and continued inclusion should improve as well. For this reason, data were collected on a peer's behavior to assess the collateral effect of the intervention.

Phase 1: FBA and Intervention Testing

Method

Setting

This study took place in a kindergarten classroom in a public elementary school in South Korea. The classroom consisted of 22 typically developing 5-year-old children, 1 child (Minsu) who had multiple disabilities and was fully included, and 2 additional children with disabilities from a segregated special classroom who were included for about 1.5 hr every day. Class sessions were held daily from 9 a.m. until 12:30 p.m.

The typical classroom schedule consisted of teacher-directed circle time; center activities; snack time; outdoor play; other teacher-directed group activities such as art, music, or science; and lunch. The classroom staff consisted of one certified early childhood education (ECE) teacher and a classroom aide. When the two partially included children participated, an early childhood special education (ECSE) teacher or a special classroom aide joined the classroom activities.

Participants

The primary participant was Minsu, a 6-year-old child who had been diagnosed with both mental retardation and autism. He had received educational services in the kindergarten classroom for 2 years because of his severely delayed developmental skills. He was formally placed in the ECSE classroom for children with disabilities between the ages of 3 and 6 years but was generally being served in the general education kindergarten classroom. The ECSE staff was supporting Minsu in the form of consulting with the ECE staff on instructional planning and implementation while limiting the one-on-one instructional time with Minsu in the special classroom. However, Minsu was not meaningfully included with typically developing peers because of limited skills and persistent challenging behavior. In class, Minsu engaged in disruptive behaviors that included crying, vocalizing, screaming, lying down on the floor, jumping, and hitting objects against the floor. He also banged his head and was aggressive toward peers by biting, hitting, scratching, and pinching. Minsu used only a few spoken words, such as *milk*, *drink*, and *mouth* to ask for food. He seldom responded when others attempted to interact. He also showed interest in only a few activities, which included swinging, performing physical movement on a therapeutic ball, and washing his face.

On the *Korean Social Maturation Scale* (Kim & Kim, 1995), Minsu obtained a Social Quotient of 57. At

the beginning of the school year, he was unable to pass most of the developmental milestones for 1- or 2-year-olds in all developmental domains. His formal education had not started until age 5, when he was accepted into the ECSE classroom. Minsu received speech therapy for 2 hr per week. He was able to eat by himself but ate only favorite foods such as meat, spicy foods, and biscuits. He needed help with dressing and using the toilet.

A second participant was Hyungjun, a typically developing 5-year-old peer who was selected because of frequent negative interactions with Minsu, which included being aggressive, grabbing objects, and pushing. Hyungjun's level of understanding peers with disabilities was low even though he had an older sister who had Down syndrome.

The ECE teacher had 5 years of teaching experience but only 1 year of experience with inclusive education. She willingly included children with disabilities and worked collaboratively with the ECSE teacher to plan lessons and implement curricula. However, she expressed difficulty in managing challenging behavior and wanted to improve her skills in this area. Her typical ways of intervening when challenging behaviors occurred was to hold the child to prevent disturbing others, isolate the child from the group and provide individual activities with ECSE staff, put the child in a quiet chair, or reprimand.

Behavioral Definitions

Challenging, appropriate, and replacement behaviors were identified for Minsu. Positive interaction behaviors were identified for the peer (Hyungjun) and for the ECE teacher. Minsu's challenging behavior was observed during both the assessment and intervention phases, but his appropriate and replacement behaviors were observed during the intervention phase only. The positive interaction behaviors of Hyungjun and the teacher were targeted for intervention. The topographies and definitions of each of these behaviors appear in Table 1.

Procedure

An FBA was conducted to identify the variables that occasioned and maintained Minsu's challenging behaviors. The assessment involved conducting structured interviews with staff, followed by collecting direct observational data in the classroom and then developing hypotheses about the apparent controlling variables. The hypotheses were then tested experimentally in the classroom during typical routines.

Structured Interviews. Interviews were conducted jointly with the child's mother and ECE and ECSE staff. One of the researchers (Gilsoon Jung) interviewed them as a group for about 2 hr. The interviews followed the *Preliminary Functional Assessment Survey* (Dunlap et al., 1993), a 22-item questionnaire designed to gather information about challenging behavior in classroom situations. Specific questions/items focus on identifying the conditions under which a target behavior is very likely or very unlikely to occur. It also includes items that identify when and how often the behavior occurs and whether its occurrence could be related to skill deficits (e.g., limited communication skills), medication, deprivation conditions (e.g., thirst, hunger), or discomfort. The FBA mainly focused on the events that triggered and maintained Minsu's problem behaviors in the general classroom. However, it also gathered information on the patterns of Minsu's behavior during situations at home and during interaction with ECSE staff to develop a broader understanding of his problem behaviors.

Structured Observations. A–B–C data (Bijou, Peterson, & Ault, 1968) were collected to corroborate the information reported during the interviews. During the A–B–C data collection, an observer (Gilsoon Jung) recorded the antecedent conditions that preceded each occurrence of a problem behavior and the consequences that followed each occurrence. Five 15-min observations, one or two observations each day for a period of a week, took place during teacher-directed group activities (i.e., circle time, music, art, and science) because the interviews had indicated that these were the times when challenging behaviors were most likely to occur. When sessions were conducted on the same day, sessions were separated by a break of at least 1 hr. The researchers discussed the results of the observations while viewing some of the video sessions to confirm the functions of Minsu's problem behaviors.

Hypothesis Development. The interview results and A–B–C data indicated that preference for certain activities, size of group, and attention from staff were related to occurrences of Minsu's challenging behavior. Interviews with Minsu's mother indicated that his problem behavior occurred most often when there was demand for engagement in nonpreferred activities and when he was not allowed to play at activities he wanted. Minsu tried to avoid all nonpreferred activities at home. His mother felt that his challenging behavior at school was related to classroom activities.

The staff also indicated that attention from them and the use of large-group activities were related to Minsu's challenging behaviors. They reported that Minsu could not independently engage in activities when in the general classroom. He had difficulties participating in group activities with peers. The ECE staff suggested that Minsu needed individual attention because the noise level in the general classroom was higher than in the special classroom, where only five children were served.

TABLE 1. Topographies and Definitions of Target Behaviors

Participant	Topography	Definition
Minsu	**Challenging behavior**	
	Being out of seat	Leaving his seat, being out of activities and wandering around the room, or attempting to get out of his seat
	Hitting himself	Hitting his head or face using his palm, fist, or objects
	Pinching	Pinching his peers or teachers using his fingers or entire hands
	Scratching	Scratching his peers or teachers using his fingernails
	Biting	Biting, or attempting to bite, his peers' or teachers' hands or shoulders with his mouth
	Crying	Crying loudly or whining (sobbing)
	Vocalizing	Making meaningless sounds, such as *ah, woo,* and *ae*
	Inappropriately using objects	Pounding blocks with hands, books, or other small play materials while sitting or moving around the classroom
	Appropriate behavior	
	Engaging	Holding peer's hands, watching peer movement, or imitating body movement (during music activities); interacting with activity materials or receiving play materials offered by peers (during center activities); looking at instructional materials or teacher, or imitating body movement while remaining in his area (during circle time)
	Following directions	Complying with teacher requests, such as imitating teacher modeling, pointing at instructional materials, and picking up play materials
	Replacement skill	
	Using communication skill	Giving a picture communication card to the teacher that shows a child and adult playing together
Hyungjun	**Positive interaction**	
	Playing with Minsu	Allowing Minsu to join his activities, sharing play materials, and helping him with manipulation of toys
	Showing affective behavior toward Minsu	Holding Minsu's hands or shoulders
	Verbally initiating interaction	Greeting Minsu or inviting him to play (e.g., "Want to do it with me?" "Want to go with me?")
Teacher	**Positive interaction**	
	Facilitating Minsu's activities and social interaction	Modeling manipulation of play materials, requesting Minsu to act on them, prompting him to respond to peers' initiation of social interaction
	Providing attention	Calling Minsu's name, gently tapping his back, conversing, or helping him with activities when he engages in the activities and when he responds to peers' initiation for social interaction

They also reported that most of the group activities did not reflect Minsu's developmental levels. They believed he did not have sufficient skills to participate in activities and interact socially. However, the ECSE staff indicated that Minsu engaged in challenging behaviors to gain attention from them when he was included in the general classroom and when he participated in activities in the special classroom. They believed he used challenging behaviors to get attention because he lacked adequate skills to express his needs.

The A–B–C data corroborated the mother's and staff's reports with regard to preferred activities and attention. Minsu engaged in appropriate behavior when he was engaged in preferred activities. He also engaged in less challenging behavior during center activities. In particular, he appropriately engaged in musical instrument play, computer games, and Play-Doh activities. However, inclusion in teacher-directed whole-group activities almost always resulted in challenging behavior that, in turn, often resulted in gaining access to teacher attention or preferred activities. During group activities, the special class staff always tried to stay with Minsu by hugging and talking to him to prevent occurrences of challenging behavior. They also allowed him to engage in individual preferred activities. However, it was not clear whether the size of the group during teacher-directed activities would change the level of challenging behavior.

Based on the results of the interviews and A–B–C data, the researchers developed the following three hypotheses: Minsu will engage in low levels of challenging behavior during group activities when (a) activities incorporate his preference or interests, (b) instruction is provided for a replacement skill, and (c) the group size is reduced.

Experimental Analysis. The teaching staff and researchers jointly developed procedures to test the hypotheses within the context of the typical classroom activities or routines. The hypotheses were tested during eight different experimental conditions using an alternating treatment design. The nature and order of the assessment conditions appear in Figure 1. Each condition (1–8) was presented in order, and the entire sequence was then repeated two more times. Thus, a total of 24 conditions were employed to test the hypotheses. All assessment conditions were tested during circle-time activities, because this was the situation in which Minsu's challenging behavior occurred most often. The hypotheses were tested for a period of 2 weeks. Each assessment session was 10 min in duration. The circle time normally lasted 20 to 25 min, and two sessions per circle time were conducted each day (10 sessions per week). A 2-min break occurred after each 10-min session. However, four sessions were conducted during the last 2 days of the second week, because circle time was held twice (morning and afternoon) on each of these days.

Implementing the assessment conditions required prior identification of preferred and nonpreferred activities for Minsu. The preference assessment was conducted using criteria identified by Dyer (1987, 1989). First, the teaching staff was asked to identify activities that Minsu would (a) initiate independently or when first prompted, (b) continue performing for several minutes, and (c) resist ending (i.e., by objecting in some way) when the activity was terminated. Using these criteria, staff identified several activities they believed would be preferred and nonpreferred. These activities then were tested directly using a 3-point scoring system (cf. Blair et al., 1999; Umbreit & Blair, 1996). When a teacher presented one of the activities, one point was scored for each of the following: beginning the activity independently or when first prompted, engaging in the activity continuously for 2 min, and resisting or objecting in any way when the activity was terminated (after 2 min). Each activity was tested within the context of naturally occurring ongoing situations. Higher scores indicated higher preference. The mean score for the preferred activities was 2.8. The mean score for the nonpreferred activities was 0.2.

During the group modification condition, staff divided the children into two groups. One group participated in the circle-time activity with the ECE staff. The other group participated in teacher-directed activities in the special classroom with the ECSE staff. Minsu participated in the activities in the general classroom.

During the replacement skill instruction condition, the ECSE staff taught Minsu to request a joint activity with staff using a picture communication card. A coated 5-cm × 4-cm photograph was made and attached to a card using Velcro. The photograph had a picture of Minsu and staff playing together. To teach the replacement skill, staff used direct, systematic procedures that included physical prompting, modeling, and time delay. Staff used hand-over-hand physical prompting while providing verbal direction, such as "Say 'Play with me.'" During the activity, the staff briefly provided Minsu with attention by helping him with his body movement or holding his hands whenever he responded to the staff's prompts by giving the communication card to them. Staff prompted use of the card whenever Minsu exhibited a precursor behavior (e.g., lifting his body, moving his head, or making a sound) for engaging in challenging behavior.

Data Collection

During each assessment condition, two independent observers (a researcher and an undergraduate student) recorded data on the target behaviors using a 10-s partial-interval recording method. The undergraduate student was trained to use the partial-interval system and practiced scoring target behaviors until a minimum of 90% interobserver agreement (IOA) was obtained. Observers scored target behaviors from live observations or videotaped recordings. About 30% of the sessions were video-

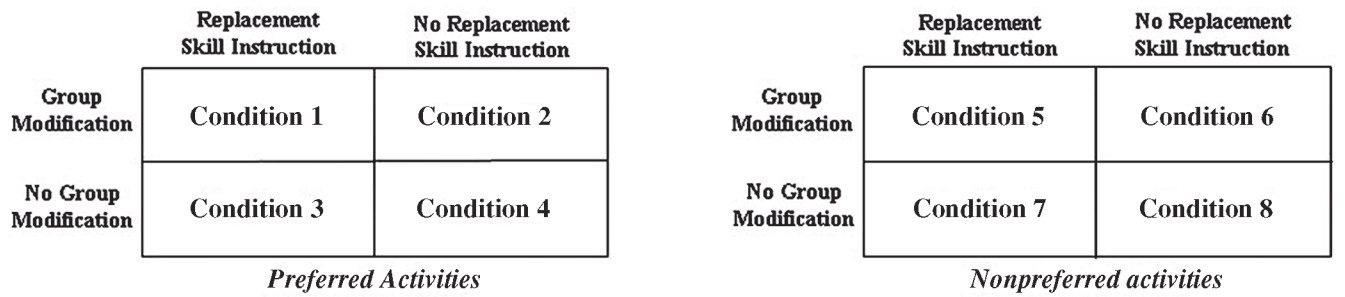


FIGURE 1. Conditions of experimental analysis.

taped for the purpose of assessing IOA. At the end of each interval, the observers recorded whether any of Minsu's challenging behaviors had occurred during the interval. Intervals scored identically by both observers were considered agreements. IOA was calculated using an exact interval-by-interval method (Kazdin, 1982) by dividing the total number of agreements by the total number of agreements plus disagreements and multiplying by 100%. Agreement averaged 92% (range = 87%–96%). Treatment integrity data were obtained after each experimental analysis condition. At the end of each session, each observer used a 3-point rating scale (2 = *well done*, 1 = *partially done*, 0 = *not at all*) for each assessment component required during the condition. All assessment components received perfect implementation scores. IOA for the treatment integrity ratings was 100% for all sessions.

RESULTS

Figure 2 presents the results of the experimental analysis for Minsu. Data in the upper left-hand panel show that the lowest levels of challenging behavior (range = 23%–

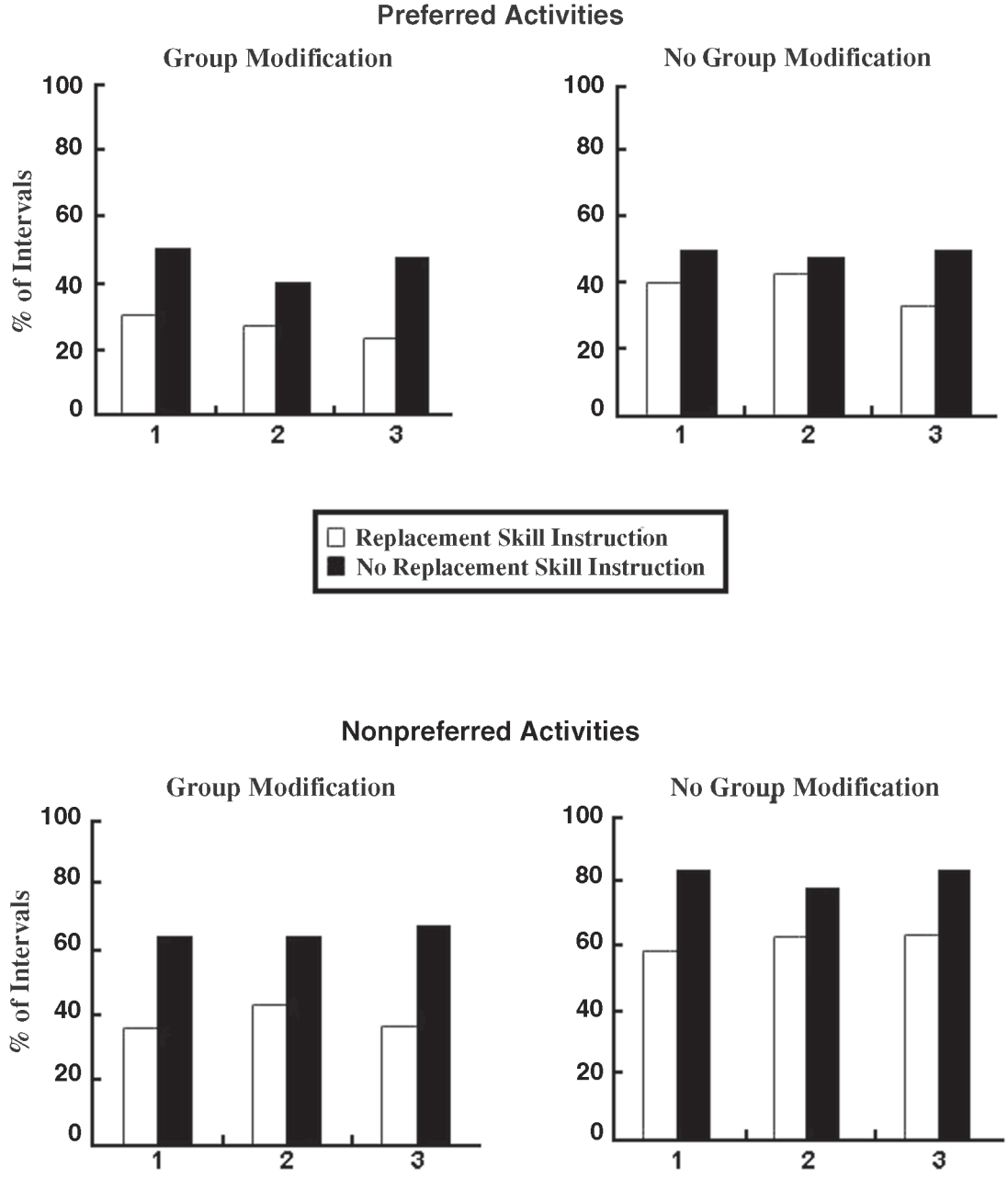


FIGURE 2. Percentages of challenging behavior during experimental analysis.

30%) occurred when Minsu participated in preferred activities, was in a smaller group, and received replacement skill instruction. Conversely, data in the lower right-hand panel show that the highest levels of challenging behavior (range = 80%–85%) occurred consistently when Minsu was provided nonpreferred activities, was in a larger group, and received no replacement skill instruction. All other combinations of conditions produced levels of challenging behavior that fell somewhere between these two extremes. Thus, the results confirmed all three hypotheses.

Phase 2: Function-Based Intervention

Method

Setting, Participants, and Behavioral Definitions

See the Phase 1 sections for this information.

Procedure

The researchers and teaching staff collaborated to develop a function-based intervention. Before they finalized the intervention strategies, two of the researchers facilitated brainstorming with the teaching staff to identify strategies that would be effective in supporting Minsu. The researchers helped the staff develop strategies based on environmental events, Minsu's developmental characteristics, and functions of the problem behavior. Analysis of the ECE classroom environment revealed that several antecedent and consequent modifications were needed. All environmental modifications were designed to increase the likelihood that Minsu would engage in appropriate and replacement behaviors rather than challenging behavior. Also, the intervention was designed to increase social interactions between Minsu and his peers and teachers. All consequent modifications were designed to provide positive reinforcement for the replacement and appropriate behaviors and to reduce the likelihood that inappropriate behaviors would be reinforced. The intervention consisted of modification of routines, replacement skill instruction, and effective ways to respond to challenging behavior. The staff participated in 1 hr and 20 min of staff training that the researchers provided each day for 3 days before implementing the intervention.

Staff Training. Training focused on how to modify activities and routines, systematically provide physical and verbal prompts, use modeling procedures to teach the communicative replacement skill, and respond effectively to occurrences of challenging behavior. Even though the ECE staff was responsible for implementing the intervention strategies, both the ECE and ECSE staff participated in the training so that they could support Minsu in a collaborative fashion in different situations. The ECSE staff, in particular the head teacher, had better knowledge of and skills in managing Minsu's behavior but were reluctant to support ECE staff with the approaches of function-based intervention. During the training, the staff learned the importance of antecedent manipulations to prevent problem behavior, how to teach and reinforce new skills, and systematic ways to extinguish problem behavior. Instruction involved videotapes in which one of the researchers demonstrated systematic direct procedures to teach Minsu the communicative replacement skill. Staff practiced the instructional methods during the training sessions and received feedback from the researcher.

Modification of Routines. Routines were modified to reduce the likelihood Minsu would engage in challenging behavior in the inclusive classroom. Opportunities to make choices were enhanced by providing preference-based materials in centers. Various responsive puzzles and picture-matching games were placed in the manipulative center. Preferred musical instruments (e.g., drums) were made available in the music center. For computer activities, educational computer games that reflected Minsu's developmental level were identified. For the art center, clay activity was introduced to incorporate Minsu's preference for motor activities, such as pulling and pounding. For his least favorite activity, the role-play center, firefighter play materials were introduced because of Minsu's interest in fire trucks. New materials, such as books with photos of real objects, were added in the language center to facilitate engagement.

The staff also incorporated choice making during center activities. They required Minsu to first choose one center from his list of least favorite centers. Each time Minsu participated in one of these activities, he was allowed to choose and engage in an activity from his list of favorite centers. Also, to prevent challenging behavior, the number of children allowed at each center was reduced—only four children were allowed in each center at any given time.

For circle-time activities, the teaching staff used multimedia, such as PowerPoint materials, a projector, and visual presentation equipment, when introducing new stories, materials, and topics related to a theme of the day. Songs with finger play and physical movements were used between interactive activities. Minsu was encouraged to explore new materials. Circle time was conducted with fewer children than during baseline. Interactive activities were used to facilitate engagement, and peer modeling was used to help Minsu learn to imitate appropriate engagement and social interaction skills. Typically

developing peers were encouraged to interact with Minsu, demonstrating appropriate behavior.

Replacement Skill Instruction. Based on the experimental analysis, Minsu was systematically taught replacement skills designed to gain teacher attention. Staff used the same picture communication card used during the experimental analysis but added a string so Minsu could wear it around his neck. Staff encouraged Minsu to use the card frequently. The design of the picture card and instructions for using it were similar to methods used in the *Picture Exchange Communication System* (Bondy & Frost, 1994). Systematic physical and verbal prompts were used and then gradually faded. Staff used both physical and verbal prompts during the first 16 intervention sessions, then only verbal prompts during the next 10 sessions. During the final 10 sessions, staff presented no prompts. When Minsu used the communication card to initiate interaction with staff, staff provided attention by tapping his back, helping him with activities, or playing with him.

Response Modification. Staff ignored (i.e., withheld attention from) Minsu's challenging behaviors when they occurred. When a particular behavior stopped, staff directed Minsu to use his communication card to make a request. If Minsu responded by using the replacement skill, staff provided attention.

A quiet area was designed in the classroom. A partition was placed around a small children's sofa to provide privacy. When Minsu engaged in an activity for 10 min without challenging behavior, he was allowed to take a break with staff for 3 min. A reinforcement delay procedure was introduced when Minsu showed initial mastery of the replacement skill. Using a time-delay procedure, staff gradually increased the time between use of the replacement behavior and reinforcement. The goal was to teach Minsu to tolerate reinforcement delay.

The researchers worked indirectly with ECE staff during the intervention phase and provided technical assistance to ECSE staff to facilitate active collaboration between ECE and ECSE staff and help with monitoring intervention implementation fidelity. During the first week of the intervention phase and 2 days before the second and third phases, the researchers provided technical assistance to ECSE staff to assist them in modeling and coaching the ECE staff with strategies for interacting with Minsu and to ensure proper implementation of the intervention. Minsu's mother did not participate in the staff training but observed him in his classroom during interactions with his teachers. The purpose was to learn strategies that she could use at home. The intervention was not formally implemented in the home, but the family was encouraged to interact with Minsu using some of the strategies developed for the classroom situations. Table 2 presents descriptions of the roles of each team member during intervention.

Design

A multiple baseline design across activities (music, center activities, and circle time) was used to evaluate the intervention's effects. During baseline, Minsu participated with peers in typical classroom activities. The ECSE staff

TABLE 2. Descriptions of the Roles of Team Members During the Intervention

Member	Role description
Mother	Participated in the process of developing intervention. Observed teachers' and Minsu's interactions in class. Supported Minsu at home, interacting with him with some of the intervention strategies developed for the classroom.
ECE teacher	Participated as an active interventionist, being responsible for implementing intervention strategies. Provided naturalistic opportunities to Minsu, practicing new skills and interacting with him individually during activities with systematic prompting and response strategies. Received training from researchers and technical assistance from ECSE teacher for ongoing support and supervision.
ECSE teacher	Helped ECE staff develop and implement intervention, providing technical assistance to ECE staff, modeling strategies, and coaching the staff. Monitored target behaviors of Minsu and peer. Monitored intervention implementation fidelity. Received training and technical assistance from researchers.
ECSE teacher aide	Helped ECE staff develop and implement intervention. Assisted ECSE teacher as needed while interacting with Minsu during activities and monitoring intervention implementation fidelity. Received training and technical assistance from researchers.
Researchers	Guided staff to develop intervention strategies based on the assessment results. Provided training to both ECE and ECSE staff and technical assistance to ECSE teacher to help with consultation for ECE staff. Monitored target behaviors of participants with ECSE teacher. Checked the reliability of intervention implementation fidelity.

Note. ECE = early childhood education; ECSE = early childhood special education.

sat behind him and responded to occurrences of challenging behavior by holding him, taking him out of the group for individual activities, or putting him in a quiet chair. Intervention was implemented in music activities starting with Session 5, in center activities starting with Session 7, and during circle time starting in Session 9.

Data Collection

The procedures used to collect data and assess IOA were identical to those used in Phase 1. In addition, data were collected on Minsu's use of appropriate behavior such as engaging in activities and following teacher directions during each routine to determine whether the intervention played a positive role in increasing the targeted appropriate behavior. Data were also collected to assess the impact of the intervention on positive interactions by the teacher and the peer (Hyungjun). Peer behavior was observed to determine whether the intervention procedures designed to facilitate positive social interaction among children would increase a nontargeted peer's positive interaction with Minsu. The teacher's behavior was also observed to determine whether the intervention procedures would increase her level of positive interaction, such as prompting Minsu's interaction with peers and materials and providing positive feedback on his appropriate behavior.

Intervention data were collected three times per week for a total of 39 sessions (i.e., 13 weeks). Follow-up data were collected once per week for 6 additional weeks. Each observation session lasted 10 min.

IOA data were collected in 30% of all sessions using the same procedures described in Phase 1. IOA averaged 91% for challenging behavior (range = 82%–97%), 91% for replacement behavior (range = 85%–93%), 94% for appropriate behavior (range = 88%–100%), 91% for peer's positive interactions (range = 82%– 97%), and 92% for teacher's positive interactions (range = 82%– 95%).

Treatment Integrity

Treatment integrity data were collected during 25% of the intervention sessions to assess the degree to which intervention components were implemented with integrity. The ECSE teaching staff (the ECSE teacher or aide who was supporting Minsu during inclusive classroom activities) and researchers jointly collected treatment integrity data across the activities using a 10-item checklist. After each session, observers independently assessed whether the ECE staff (the general classroom teacher and classroom aide) did the following:

1. discussed the classroom lesson plan with the ECSE staff,
2. modified activities based on Minsu's preferences and interests,
3. modified the group size,
4. provided peer proximity and encouraged modeling of appropriate behavior,
5. provided choices to Minsu,
6. taught the replacement skill using the systematic direct teaching procedures (e.g., physical and verbal prompts during the first stage of intervention, verbal prompts during the second stage of intervention, and no prompts during the third stage of intervention),
7. responded to Minsu's precursor behaviors by prompting for the replacement skill,
8. provided attention when Minsu responded appropriately to prompts or initiated the behavior without prompts,
9. ignored challenging behavior during activities, and
10. reinforced Minsu's other appropriate behaviors.

Each item was scored using a 3-point rating scale (2 = *well done,* 1 = *partially done,* 0 = *not at all*). Both the ECE teacher and classroom aide received ratings of 1.7, with 2 being the highest possible score. IOA for treatment integrity, calculated using the point-by-point method (item by item; Kazdin, 1982) was 96% for the teacher and 92% for the aide.

Social Validity

To assess the acceptability and feasibility of the function-based intervention, ECE staff (the general classroom teacher and classroom aide) completed a 5-item social validity questionnaire. Items targeted the usability of the function-based intervention, the feasibility of teaching the replacement skill, satisfaction with the overall intervention procedures and results, and the possibility of continued future usage of the approach. Each response used a 5-point rating scale (5 = *strongly agree*, 1 = *strongly disagree*).

Results

Figure 3 shows the levels of Minsu's target behaviors during each classroom routine. Minsu engaged in high levels of challenging behavior across the routines during baseline and low levels of challenging behavior during intervention. During baseline, the mean percentage of occurrences of challenging behavior during music, center activities, and circle time was 85% (range = 82%–88%), 83% (range = 80%–85%), and 87% (range = 75%–93%), respectively. During intervention, the mean percentage of occurrences of challenging behavior was 23% (range =

10%–42%) during music, 20% (range = 10%–37%) during center activities, and 21% (range = 11%–35%) during circle time.

Figure 3 also shows the level of replacement skill use during each routine. During baseline, no replacement skill use was observed. However, during intervention, Minsu used the replacement skill to gain access to attention and tangibles. The mean percentage of intervals for replacement skill use during intervention for music, center activities, and circle time was 18% (range = 10%–23%), 14% (range = 11%–18%), and 16% (range = 12%–227%), respectively. During the first phase of intervention, when the staff implemented the intervention procedures with physical and verbal prompts, occurrences of the replacement skill increased to 13% (range = 10%–17%) during music, to 16% (range = 13%–18%) during center activities, and to 19% (range = 18%–22%) during circle time. During the second phase of intervention, when only verbal prompting was provided, use of the replacement skill continued. The mean percentage of replacement skill use during Phase 2 was 13% (range = 10%–17%) for music, 13% (11%–15%) for center activities, and 14% (range = 11%–17%) for circle time. During Phase 3, when staff employed no prompt procedure, Minsu continued to use the replacement skill at these levels.

A similar pattern was observed for appropriate behavior. Probe data on appropriate behavior across routines showed that Minsu's appropriate behavior increased during intervention. As shown in Figure 3, appropriate behavior was observed during an average of only 3% of the intervals during baseline in music. During interven-

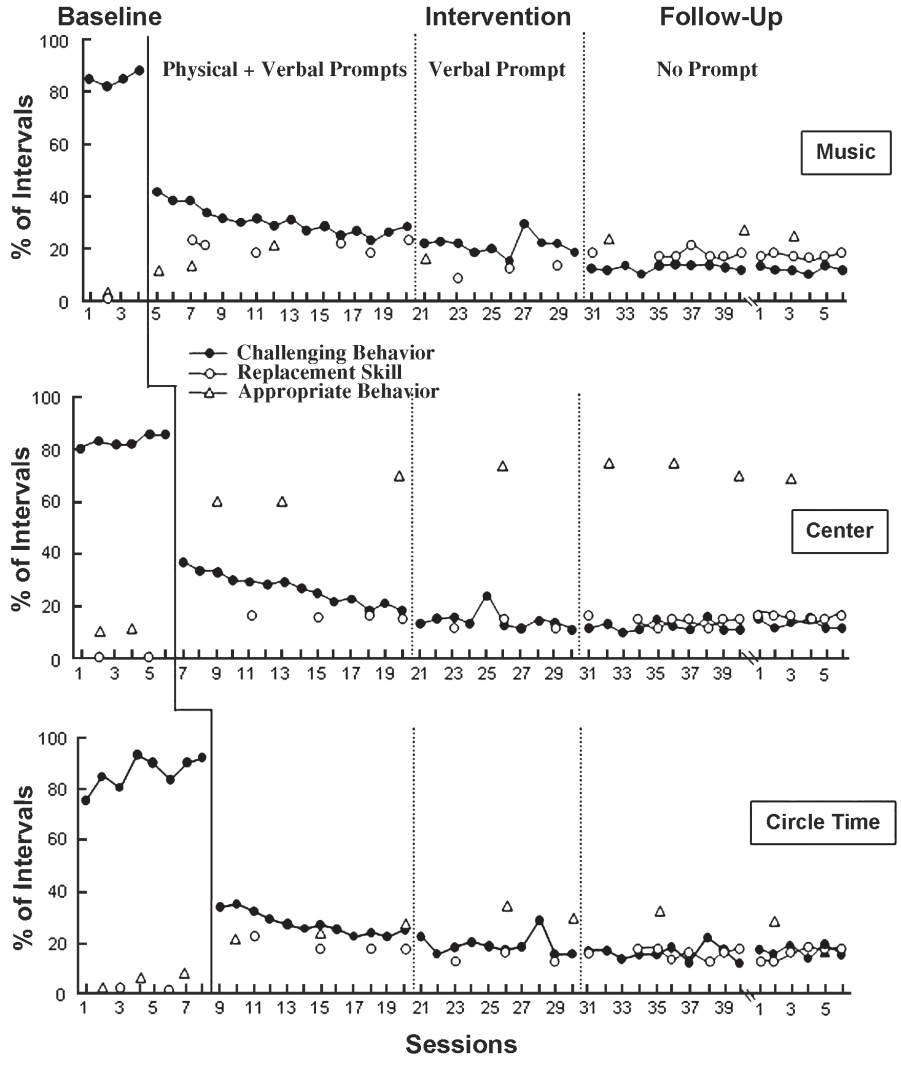


FIGURE 3. Percentages of replacement skills, appropriate behavior, and challenging behavior during baseline and intervention.

tion, these behaviors during music were observed during an average of 20% of the intervals. During center activities, in which interactions with peers and materials were targeted, the level of appropriate behavior increased from 13% (range = 12%–14%) during baseline to 73% (range = 60%–77%) during intervention. Improvements also occurred during circle time, as the level of appropriate behavior increased from an average of 7% (range = 2%–14%) of the intervals during baseline to an average of 27% (range = 22%–33%) during intervention. Follow-up data indicated that the decreased levels of challenging behavior and increased levels of appropriate behavior continued after intervention.

Figure 4 shows levels of positive peer and teacher interactions across routines. Hyungjun engaged in low levels of positive interaction with Minsu during baseline and much higher levels of positive interaction during intervention. During baseline, the mean percentages of positive interaction were 5% (range = 3%–8%), 8% (range = 5%–12%), and 5% (range = 3%–7%) during music, center activities, and circle time, respectively. During intervention, mean percentages increased to 38% (range = 22%–50%) during music, 62% (range = 32%–80%) during center activities, and 46% (range = 20%–57%) during circle time. Hyungjun interacted with Minsu more and more as the intervention progressed.

A similar pattern was observed for teacher behavior. During baseline, the mean percentage of intervals with positive teacher interaction was 21% (range = 17%–25%) during music, 23% (range = 17%–28%) during center activities, and 15% (range = 7%–22%) during circle time. During intervention, it increased to 61% (range = 53%–66%) during music, 69% (range = 60%–77%) during center activities, and 64% (range = 53%–75%) during circle time. Increased levels of positive interaction by both Hyungjun and the teacher continued during the follow-up phase.

Ratings on the social validity questionnaire indicated strong support for the function-based intervention. Scores averaged 4.5 (out of 5), and all individual item responses received a rating of 4 or 5.

Discussion

Several key findings emerge from this experimental single-case analysis. First, experimental analyses conducted in natural classroom contexts validated the hypotheses gen-

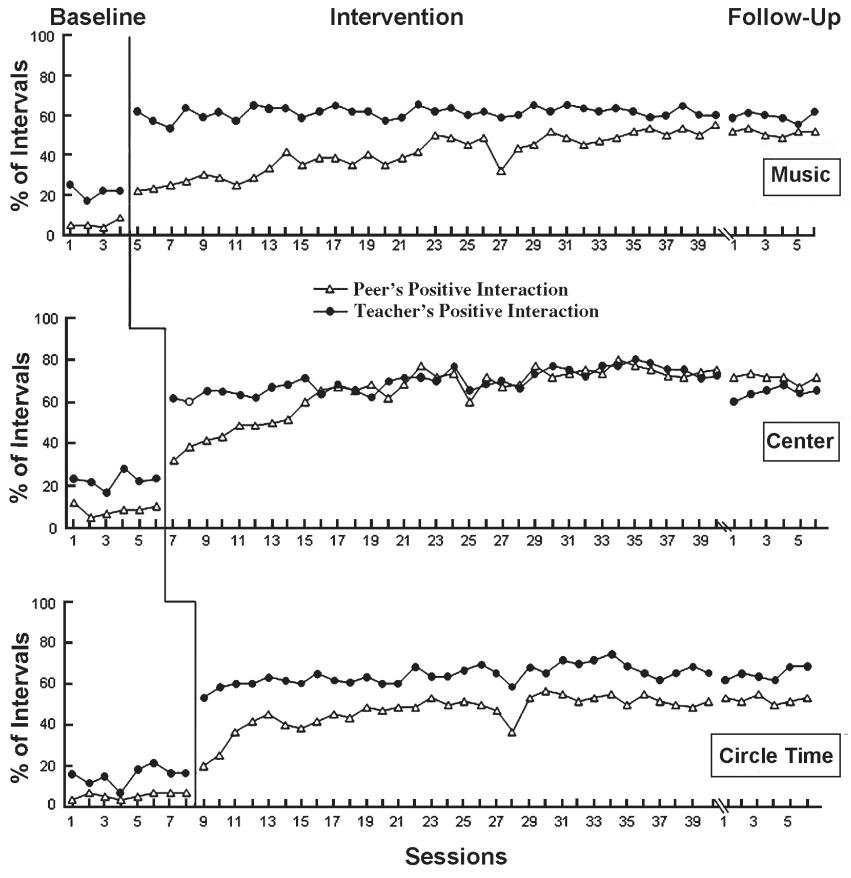


FIGURE 4. Percentages of positive interaction by peer and teacher across activities.

erated by an FBA team from structured interviews and descriptive data collection. Second, the multicomponent intervention derived from the FBA process was effective in reducing Minsu's challenging behaviors and increasing his appropriate behaviors and use of a targeted replacement skill. Third, collateral findings were markedly increased rates of positive interactions directed toward Minsu by a designated peer (Hyungjun) and by Minsu's teacher. Fourth, data on treatment integrity indicated that the intervention's implementation was high. Fifth, responses by classroom staff on the social validity assessment clearly indicated staff regarded the intervention as feasible and effective. Finally, and perhaps most important, the intervention enabled Minsu to continue participating in the inclusive educational setting (which was jeopardized due to his high baseline levels of challenging behaviors) and to exhibit generally improved patterns of behavior that maintained over extended periods of time. Given the widely acknowledged importance of social inclusion and the steadily increasing emphasis on providing early intervention supports in natural environments, the significance of these data have generality beyond the obvious benefits that affected Minsu.

The data from this study serve to replicate and extend contributions from earlier investigations. From the perspective of functional assessment, the current findings echo numerous predecessors in which rigorous procedures, including functional analysis, have been used to empirically confirm hypotheses derived from indirect and solely descriptive data collection procedures (e.g., Dunlap, Kern-Dunlap, Clarke, & Robbins, 1991; Kern, Childs, Dunlap, Clarke, & Falk, 1994; Umbreit & Blair, 1996). Such data indicate that analog contexts are not necessary to establish or validate the presence of functional relations governing the occurrence of challenging behaviors. From the perspective of assessment-based intervention, the data from this study replicate earlier reports in which researchers used functional assessment information to construct individualized interventions demonstrated with experimental procedures to be effective in reducing problems and increasing desired behaviors (e.g., Blair et al., 1999; Conroy et al., 2005; Duda et al., 2004; Dunlap et al., 1991; Lewis & Sugai, 1996). The present investigation contributes evidence to a growing body of literature that clearly attests to the benefits of using functional assessment information to inform the development of behavior support interventions (Dunlap & Carr, 2007), including interventions for young children with disabilities (e.g., Conroy et al., 2005).

The experimental analysis of the assessment-based intervention, depicted in Figure 3, shows an immediate and considerable effect on challenging behavior. In all three contexts, the initial reduction was greater than 50%, and the decreases continued throughout the duration of data collection. Although the effects were also conspicuous for appropriate and replacement behaviors, the magnitude of change for these dependent measures was smaller. This may be explained by a relative lack of opportunity in most contexts for Minsu to engage in appropriate and replacement behaviors, as defined for this study (see Table 1). Under naturally occurring circumstances, one would not expect a child to communicate a particular request (the replacement behavior) during a high percentage of intervals. Nevertheless, it is encouraging that the levels during all phases of intervention were clearly higher than during baseline. For appropriate behavior, Minsu's responding conformed to the definition during center activities for a high proportion of the observed intervals. His behavior during music and circle time did not correspond as frequently to these context-specific operational definitions (see Table 1). Although improvement relative to the baseline was obvious, the extent to which Minsu's levels of appropriate behavior matched the levels of his classroom peers is unknown. It would have been interesting and informative to collect such normative data. However, such data collection was beyond the scope of the current investigation.

An important and encouraging result of the experiment was the increased levels of positive interactions directed toward Minsu by his teacher and by his classmate Hyungjun. These data represent an extension beyond existing research, as previous studies have not reported such collateral findings in the current form. Increased positive interactions occurred immediately following intervention in each context and maintained across the study. An explanation for the increase in the teacher's positive interactions can be traced to several of the intervention components, in particular those that required increased rates of prompting, delivery of choice opportunities, and reinforcement. An explanation for Hyungjun's increase is not so apparent. However, it was likely a function of the teacher's modeling and prompting of peer proximity, as well as the overall improvement in Minsu's behavior.

It is important to acknowledge some limitations of the investigation. First, and most obviously, the analysis was restricted to one classroom and one individual child with disabilities. This limitation might be mitigated to some extent because there is a large amount of data with this one case and because the data are complemented by numerous predecessors for which the current study is a systematic replication. However, generality remains a concern because there were undoubtedly characteristics of the participating classroom that are not present in every context. The classroom personnel were willing to engage in a collaborative and somewhat effortful process of assessment, intervention, and environmental reengineering that staff in other classrooms might resist. Indeed, the level of cooperation and commitment might have been unusually high, making it appear as if research

into such variables as investment, dedication, and cooperation could be a fruitful direction to pursue (cf. Hieneman & Dunlap, 2000). Finally, it is noteworthy that the study took place in South Korea. The vast majority of research in this area has occurred in the United States. Data from other countries carry questions regarding the possibility of cultural influences and the potential impact of different policies and attitudes. Although the congruence of the current data with so many published reports from the United States suggests strong external validity, other questions about cross-cultural patterns await exploration and examination.

The results of the study suggest that the function-based intervention approach is culturally relevant in supporting children with disabilities and challenging behavior in inclusive settings in South Korea. Factors that indicate such relevance include a growing awareness that problem behavior can be addressed in a positive manner (Blair & Lee, 2005; Lee & Blair, 2005; S. Park et al., 2001), an emphasis on general educator and family involvement in the process of Individualized Education Program development and implementation (Blair, Cheon, Park, & Hong, 2005), and a heightened willingness of general educators to support students with disabilities and challenging behavior in inclusive classrooms. One implication for practice and policy development in South Korea is the emphasis on training and technical assistance for the teaching staff during the process of function-based intervention. As of 2005, 51% of schools across the nation had banned corporal punishment themselves, nearly double the percentage in 2003 (27.7%; "Prohibition of Corporal Punishment in Schools," 2006). Accordingly, the Korea Ministry of Education and Human Resources Development has been working on establishing a legal basis for prohibiting corporal punishment that infringes on human rights ("Calls for a Ban on Corporal Punishment," 2006). Although this recent change in policies and practices regarding discipline of students has been demanding teachers to build positive relationships with students, many general educators describe themselves as lacking the skills needed to address the problem behaviors of students in positive ways ("Prohibition of Corporal Punishment in Schools," 2006). Therefore, training and technical assistance for the teaching staff during the process of function-based intervention should focus on collaboration among general and special educators to support children with disabilities and challenging behavior. This year, Korea enacted the Individuals With Disabilities Special Education Act that replaced the existing Special Education Promotion Act (Korea Ministry of Education and Human Resources Development, 2007), which ensures the support of these students while requiring collaboration between general and special educators in the Individualized Education Program process. Developing behavior support plans should be one of the most important areas of the collaboration to help with the successful inclusion of students with disabilities.

In summary, the results of the present study provide additional testimony to the efficacy of functional assessment and assessment-based interventions for young children with disabilities. The impact of this approach was demonstrated in an effort to promote the continuation of an inclusive placement jeopardized by serious challenging behaviors. The findings are highly encouraging, not only because they add to the growing literature on assessment-based behavior support, but also because they indicate an effective strategy for helping young children with serious disabilities remain in supportive contexts in their natural environments. ◆

REFERENCES

Bijou, S. W., Peterson, R. F., & Ault, M. H. (1968). A method to integrate descriptive and experimental field studies at the level of data and empirical concepts. *Journal of Applied Behavior Analysis*, *1*, 175–191.

Blair, K. C., Cheon, B., Park, H., & Hong, E. (2005). An evaluation of individualized education programs in early childhood special education settings. *Korean Journal of Special Education*, *12*(1), 339–370.

Blair, K. C., & Lee, I. (2005). Current status and future directions of functional assessment-based behavior supports for young children with special needs. *Journal of Special Education: Theory and Practice*, *6*(3), 69–92.

Blair, K. C., Liaupsin, C. J., Umbreit, J., & Kweon, G. (2006). Function-based intervention to support the inclusive placements of young children in Korea. *Education and Training in Developmental Disabilities*, *41*, 48–57.

Blair, K. C., Umbreit, J., & Bos, C. S. (1999). Using functional assessment and children's preferences to improve the behavior of young children with behavioral disorders. *Behavioral Disorders*, *24*, 151–166.

Blair, K. C., Umbreit, J., & Eck, S. (2000). Analysis of multiple variables related to a young child's aggressive behavior. *Journal of Positive Behavior Interventions, 2,* 33–39.

Bondy, A. S., & Frost, L. A. (1994). The picture exchange communication system. *Focus on Autistic Behavior*, *9*(3), 1–19.

Calls for a ban on corporal punishment of Ministry of Education and Human Resource Development. (2006, August 18). Retrieved May 20, 2007, from http://www.donga.com/fbin/output?f=total&n=200608180321&top20=1

Carta, J. J., Sideridis, G., Rinkel, P., Guimaraes, S., Greenwood, C., Baggett, K., et al. (1994). Behavioral outcomes of young children prenatally exposed to illicit drugs: Review and analysis of experimental literature. *Topics in Early Childhood Special Education*, *14*, 184–216.

Conroy, M. A., Dunlap, G., Clarke, S., & Alter, P. J. (2005). A descriptive analysis of positive behavioral intervention research with younger children with challenging behavior. *Topics in Early Childhood Special Education*, *25*, 157–166.

Crone, D. A., Hawken, L. S., & Bergstrom, M. K. (2007). A demonstration of training, implementing, and using functional behavioral assessment in 10 elementary and middle school settings. *Journal of Positive Behavior Interventions*, *9*, 15–29.

Cunningham, E., & O'Neill, R. E. (2000). Comparison of results of functional assessment and analysis methods with young children

with autism. *Education and Training in Mental Retardation and Developmental Disabilities*, *35*, 406–414.

Duda, M. A., Dunlap, G., Fox, L., Lentini, R., & Clarke, S. (2004). An experimental evaluation of positive behavior support in a community preschool program. *Topics in Early Childhood Special Education*, *24*, 143–155.

Dunlap, G., & Carr, E. G. (2007). Positive behavior support and developmental disabilities: A summary and analysis of research. In S. L. Odom, R. H. Horner, M. Snell, & J. Blacher (Eds.), *Handbook of developmental disabilities* (pp. 469–482). New York: Guilford Press.

Dunlap, G., Hieneman, M., Knoster, T., Fox, L., Anderson, J., & Albin, R. W. (2000). Essential elements of inservice training in positive behavior support. *Journal of Positive Behavior Interventions*, *2*, 22–32.

Dunlap, G., Kern, L., dePerczel, M., Clarke, S., Wilson, D., Childs, K. E., et al. (1993). *Preliminary functional assessment survey.* Unpublished manuscript, University of South Florida, Division of Applied Research and Educational Services, Tampa.

Dunlap, G., Kern-Dunlap, L., Clarke, S., & Robbins, F. R. (1991). Functional assessment, curricular revision, and severe behavior problems. *Journal of Applied Behavior Analysis*, *24*, 387–397.

Dyer, K. (1987). The competition of autistic stereotyped behavior with usual and specially assessed reinforcers. *Research in Developmental Disabilities*, *8*, 607–626.

Dyer, K. (1989). The effects of preference on spontaneous verbal requests in individuals with autism. *Journal of the Association for Persons With Severe Handicaps*, *14*, 184–189.

Ellis, J., & Magee, S. K. (1999). Determination of environmental correlates of disruptive classroom behavior: Integration of functional analysis into public school assessment process. *Education & Treatment of Children*, *22*, 291–316.

Fox, J. J., & Gable, R. A. (2004). Functional behavioral assessment. In R. B. Rutherford, M. M. Quinn, & S. R. Mathur (Eds.), *Handbook of research in emotional and behavioral disorders* (pp. 143–162). New York: Guilford Press.

Fox, L., Dunlap, G., & Cushing, L. (2002). Early intervention, positive behavior support, and transition to school. *Journal of Emotional and Behavioral Disorders*, *18*, 149–157.

Hagopian, L. P., Wilson, D. M., & Wilder, D. A. (2001). Assessment and treatment of problem behavior maintained by escape from attention and access to tangible items. *Journal of Applied Behavior Analysis*, *34*, 229–232.

Heckaman, K., Conroy, M., Fox, J. J., & Chait, A. (2000). Functional assessment-based intervention research on students with or at risk for emotional and behavioral disorders. *Behavioral Disorders*, *25*, 196–210.

Hieneman, M., & Dunlap, G. (2000). Factors affecting the outcomes of community-based behavioral support: I. Identification and description of factor categories. *Journal of Positive Behavior Interventions*, *2*, 161–169.

Kazdin, A. E. (1982). *Single-case research designs.* New York: Oxford University Press.

Kern, L., Childs, K. E., Dunlap, G., Clarke, S., & Falk, G. D. (1994). Using assessment-based curricular intervention to improve the classroom behavior of a student with emotional and behavioral challenges. *Journal of Applied Behavior Analysis*, *27*, 7–19.

Kim, S., & Kim, O. (1995). *Korean social maturation scale.* Seoul, South Korea: ChungAng Aptitude.

Korea Ministry of Education and Human Resources Development. (2007). *Annual report on special education.* Seoul, South Korea: Author.

Lee, I., & Blair, K. C. (2005). Positive behavior support in multisettings for a young child with autism. *Korean Journal of Early Childhood Education*, *5*(1), 161–190.

Lewis, T. J., & Sugai, G. (1996). Descriptive and experimental analysis of teacher and peer attention and the use of assessment-based intervention to improve pro-social behavior. *Journal of Behavioral Education*, *6*, 7–24.

Palfrey, J., Singer, J., Walker, D., & Butler, J. (1987). Early identification of children's special needs: A study in five metropolitan communities. *Journal of Pediatrics*, *111*, 651–659.

Park, E., & Park, S. (2001). A survey of special education teachers' perceptions on education for students with severe disabilities. *Korean Journal of Special Education*, *36*(1), 29–55.

Park, S., Park, E., & Oh, J. (2001). An analytic review of current research practices: Interventions with problem behaviors of children with severe disabilities. *Korean Journal of Special Education*, *36*(2), 46–63.

Prohibition of corporal punishment in schools. (2006, October 20). Retrieved May 20, 2007, from http://korea.assembly.go.kr/abo/zin_read.jsp?cha=19&boarditemid=1000005519

Reichle, J., McEvoy, M., Davis, C., Rogers, E., Feeley, K., Johnston, S., et al. (1996). Coordinating pre-service and in-service training of early interventionists to serve preschoolers who engage in challenging behavior. In L. K. Koegel, R. L. Koegel, & G. Dunlap (Eds.), *Positive behavioral support: Including people with difficult behavior in the community* (pp. 227–264). Baltimore: Brookes.

Sasso, G., Conroy, M. A., Stichter, J., & Fox, J. J. (2001). Slowing down the bandwagon: The misapplication of functional assessment for students with emotional and behavioral disorders. *Behavioral Disorders*, *26*, 282–296.

Scott, T. M., & Nelson, C. M. (1999). Functional behavioral assessment: Implications for training and staff development. *Behavioral Disorders*, *24*, 249–252.

Seven in 10 schools allow corporal punishment. (2003, September 14). *The Korea Times.* Retrieved May 20, 2007, from http://www.koreatimes.co.kr/

Strain, P., & Cordisco, L. K. (1994). LEAP preschool. In S. L. Harris & J. S. Handleman (Eds.), *Preschool education for children with autism* (pp. 225–244). Austin, TX: PRO-ED.

Sugai, G., Horner, R. H., Dunlap, G., Hieneman, M., Lewis, T. J., Nelson, C. M., et al. (2000). Applying positive behavioral support and functional assessment in schools. *Journal of Positive Behavior Interventions*, *2*, 131–143.

Umbreit, J. (1996). Functional analysis of disruptive behavior in an inclusive classroom. *Journal of Early Intervention*, *20*, 18–29.

Umbreit, J., & Blair, K. (1996). The effects of preference, choice, and attention on problem behavior at school. *Education and Training in Mental Retardation and Developmental Disabilities*, *32*, 151–161.

Umbreit, J., & Blair, K. C. (1997). Using structural analysis to facilitate the treatment of noncompliance and aggression in a young child at-risk for behavioral disorders. *Behavioral Disorders*, *22*, 75–86.

Volkert, V. M., Lerman, D. C., & Vorndran, C. (2005). The effects of reinforcement magnitude on functional analysis outcomes. *Journal of Applied Behavior Analysis*, *38*, 147–162.

Whalen, C. K., & Henker, B. (1985). The social worlds of hyperactive (ADDH) children. *Clinical Psychology Review*, *5*, 447–478.